Department of Agriculture

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PHILIPPINE RICE RESEARCH INSTITUTE

2021

PHILRICE R&D HIGHLIGHTS

CROP BIOTECHNOLOGY CENTER

Content

DIVISION

Crop Biotechnology Center

Roel R. Suralta

EXECUTIVE SUMMARY

The Crop Biotechnology Center was established through Administrative Order 21 to implement a rationalized, effective, and efficient AgBiotech R&D agenda for the Department of Agriculture (DA) to generate improved agricultural technologies, productivity, and enhanced commercial potential value and crop activities. With the issuance of Administrative Order 26 series of 2021, the Center›s goal was to develop and utilize modern biotechnologies for agriculture and fisheries and extension services to research for development (R4D) network of institutions towards the development of policy, technology, and product. The Center undertakes innovative R4D programs, comprehensively covering fundamental research to innovation phases using advanced biotechnology tools, and ensuring technology transfer or commercialization of outputs generated through R4D programs.

The Center applies advanced technologies in combination with experimental and computational methods to discover, quantify and validate essential traits that increase yield, improve the quality of rice (value-adding), and enhance varietal development for grain quality and abiotic and biotic stress tolerance. The genes controlling the traits are also dissected to improve screening (phenotyping) systems and selection efficiency in crop breeding systems. The Center also performs fundamental and applied research in the areas of genome biology, computational biology, genetics, microbial systems, and molecular biology.

PROJECT 1

Candidate Gene Analysis and Molecular Marker Development for Rice Breeding

Arlen A. Dela Cruz

The project identified candidate genes influencing rice traits to increase selection efficiencies in rice varietal improvement. Functional molecular markers highly associated with plant phenotypic variation were also designed. Its component studies implement molecular genetic variation studies on targeted traits for agronomic, biotic, and abiotic stress resistance.

The project includes the following:

1. Three rice lines with functional stay-green character under drought conditions;
2. Four candidate functional stay-green QTLs;
3. Four root plasticity QTLs for improving adaptation to water stresses;
4. Two candidate QTLs associated with GLH resistance;
5. Two recombinant bacteria with antigenic tungro virus coat protein gene; and
6. Sixteen candidate genes controlling root plastic traits available from our Philippine germplasm.

QTL ANALYSIS OF ROOT PLASTICITY UNDER WATER STRESS CONDITIONS

Jonathan M. Niones, Roel R. Suralta, Antoinette S. Cruz, and Ricky Jay M. Gonzaga

As shown in Table 1, 10 QTLs associated with root plasticity with a logarithm of odds (LOD) value of greater than or equal to 2.5 were identified using 33 polymorphic markers and phenotype data from 160 RILS Nipponbare/CSSL47*2 Nipponbare.

PROJECT 1

Table 1. List of QTLs detected along Chromosome 12 on 160 RILs Nipponbare/CSSL47*2 Nipponbare under Soil Moisture Fluctuation condition.

Trait	QTL	Chromosome	Nearest marker	Position (bp)	Flanking Markers		LOD	Additive Effect	Allele	R^2
					Left	Right				
Total L-type Lateral Root Number	*qLLRN12*	12	RM27850	8,471,181	RM27847 8,145,345	RM27866 8,797,017	2.93	-13.70	Kasalath	0.0820
Total Lateral Root Number	*qTLRN12.1*	12	RM247	7,361,128	RM27793 7,132,255	RM6296 7,590,000	3.1	-38.12	Kasalath	0.0890
	qTLRN12.2	12	RM27866	8,578,607	RM27850 8,327,767	RM101 8,829,447	3.56	-41.21	Kasalath	0.1010
	qTLRN12.3	12	RM27877	9,186,893	RM3246 9,095,058	RM27879 9,278,728	3.33	-43.31	Kasalath	0.0956
	qTLRN12.4	12	RM27900	9,586,026	RM27892 9,504,613	RM27907 9,667,439	4.76	-47.28	Kasalath	0.1328
	qTLRN12.5	12	RM27920	10,135,181	RM5939 10,067,290	RM27923 10,203,072	7.36	-73.69	Kasalath	0.1995
	qTLRN12.6	12	RM27940	10,594,250	RM27928 10,320,134	RM27944 10,868,365	7.98	-84.31	Kasalath	0.2405
	qTLRN12.7	12	RM27959	11,992,933	RM27944 10,868,365	TG156 13,117,500	2.89	-59.46	Kasalath	0.1344
Total Lateral Root Length	*qTLRL12*	12	TG154	3,331,839	RM27492 1,632,808	RM27706 5,030,870	5.99	0	Kasalath/ Nipponbare	0.3639
Total Root Length	*qTLRN12.1*	12	TG154	3,331,839	RM27492 1,632,808	RM27706 5,030,870	5.67	0	Kasalath/ Nipponbare	0.3353

The target QTL, *qLLRN12*, is found between RM27847 and RM27866, with RM27850 being the QTL's closest/peak marker (Figure 1). The location is estimated to be around 8,471,181bp (30.80cM). However, compared to the prior report, the distance between flanking markers of *qLLRN12* has shortened (Niones et al., 2015). We narrowed it down from 5,527,500bp (20.1cM) to 651,672bp (2.37cM).

We will locate the candidate genes after we find this QTL. First, using sequence databases, candidate genes associated with the measured root plasticity traits will be carried out in databases. Then, candidate genes that are associated with the measured root plasticity traits will be carried out using bioinformatic servers such as the Phytozome-11 rice genome browser (https://phytozome.jgi.doe.gov/pz/portal.html) and the MSU Rice Genome Annotation Project (http://rice.plantbiology.msu.edu) and the RAP Rice Annotation Project Database (rapdb.dna.affrc.go.jp). Candidate genes will be searched within 200 kb genomic regions, i.e., and + 100,000 bp (Zhao et al. 2011). The candidate genes will be classified and annotated using the Rice Genome Annotation Project and the RAP Rice Annotation Project Database.

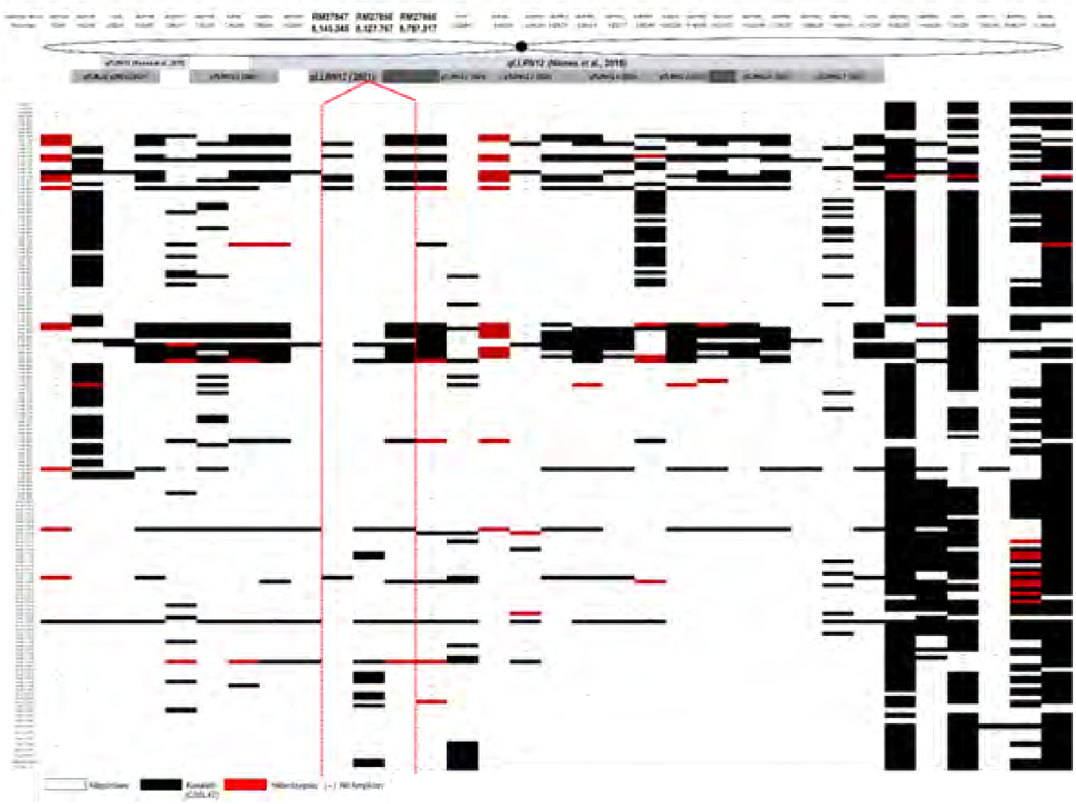

Figure 1. Graphical map of QTLs detected along Chromosome 12 on RILs of Nipponbare/ CSSL47*2Nipponbare under soil moisture fluctuation condition. The previous study detected QTLs in the light blue bar (Niones et al., 2015) while QTLs in the grey bar are newly identified. Red broken lines show the location of *qLLRN12* while patterned bars indicate that QTLs overlapped in the same region, n=162.

QTL ANALYSIS OF FUNCTIONAL STAY GREEN TRAITS UNDER WATER STRESS CONDITIONS

Roel R. Suralta, Jonathan M. Niones, Antoinette Cruz and Ricky Jay Gonzaga

Fifty-one markers distributed across the 12 chromosomes were identified between NSIC Rc 160 and Kutsiyam. Together with the phenotypic data of 275 RILs NSIC Rc 160 X Kutsiyam screened using the bucket method under progressive drought (PDR), we conducted QTL analysis using Win QTL Cartographer 2.5. Seven QTLs associated with functional stay-green traits were found using 51 SSR markers and SPAD data (Day 2 - Day 14 after heading) under drought conditions. These QTLs were discovered in the regions of Chromosome 3, Chromosome 10, and Chromosome 11, and they were attributed to five different phenotypes. Two QTLs in SPAD 2 days after heading (DAH) are located in Chromosome 3 (*qSPAD2-3*) and Chromosome 11 (*qSPAD2-11*); 1 QTL in SPAD 4DAH is located in Chromosome 3 (*qSPAD4-3*); 2 QTLs in SPAD 6DAH are located in Chromosome 10 (*qSPAD6-10*) and Chromosome 11 (*qSPAD6-11*); 1 QTL in SPAD 10DAH is located in Chromosome 11 (*qSPAD10-11*), and 1 QTL in SPAD 12DAH is located in Chromosome 11 (*qSPAD12-11*). All of the QTLs we detected except for *qSPAD6-10*, which is contributed by the other parent, NSIC Rc160 with good grain yield, are contributed by Kutsiyam, which is known to have the stay-green trait.

The QTLs found on Chromosome 3 are in the same region as on Chromosome 11 (Figure 2). QTLs in Chromosome 3 were found between RM523 and RM14566 with a distance of 4,414,468bp between the two flanking markers. In contrast, QTLs in Chromosome 11 were found between TG148 and RM209 with a distance of 7,750,862bp between the two flanking markers. QTL is also found on Chromosome 10 between RM5095 and RM25152 with a distance of 6,090,286bp. Interestingly, QTLs on Chromosome 11 are discovered in four traits, varying from 2DAH to 12DAH. It indicates that the QTL present could be linked to the functional stay-green trait. However, the distance between markers where the QTLs were found was observed. Therefore, the addition of markers in between the regions where QTLs were detected will be the next step in narrowing the distance and locating the QTLs linked to the functional stay-green trait.

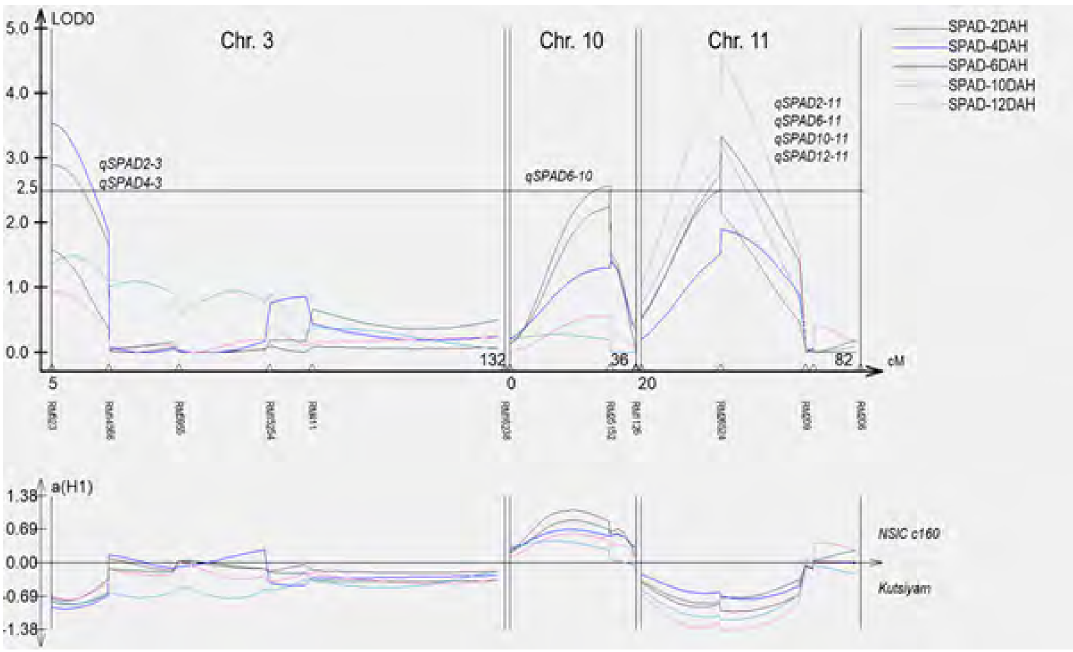

Figure 2. QTLs were detected in Chromosome 3 (Chr.3), Chromosome 10 (Chr.10), and Chromosome 11 (Chr. 11) in 275 RILs NSIC Rc 160 X Kutsiyam on SPAD value under Progressive drought at 0-14 days after heading (DAH). cM – centimorgan.

PROJECT 1

MAPPING OF NOVEL GLH RESISTANCE QTL

Arlen A. dela Cruz, Teodora E. Mananghaya, Editha V. Evangelista, Mary Joyce G. Yapchiongco, Eleanor S. Avellanoza, Ma. Johna C. Duque, Christopher C. Cabusora, and Nenita V. Desamero

In mapping novel GLH resistance QTL other than Glh14 among PTRVs, more than 100 additional SSR markers were screened for polymorphism between PTRV Lead and TN1. Among these, 83 markers appeared monomorphic, 16 were polymorphic, and the others yielded no amplification or bad amplification. Table 2 shows the number of SSR markers used to survey the different rice chromosome regions and the number of polymorphic SSR markers in every chromosome based on the surveys conducted in 2020 and 2021.

The 16 polymorphic SSR markers were identified and used to genotype the same F2 mapping population studied in the initial mapping activities last year, composed of 257 plants. The QTL and linkage analysis was conducted using ICIMapping v4.2. The LOD threshold for identifying QTL was set at 2.5. Figure 3 shows the locations of the 58 polymorphic SSR markers used in the updated QTL analysis and the positions of the QTL detected. Table 3 shows the updated QTL and linkage analysis showed a consistent association of QTL regions at chromosomes 5 and 10 with the percent nymph mortality observed on the fourth and fifth days of the antibiosis test. The computed QTL LOD values were between 4.5 to 15.6, which showed strong evidence of the presence of QTL in the regions. The percent variation observed, however, were all less than 10.0% hence associated the GLH resistance in PTRV Lead with two minor QTL in rice chromosomes 5 and 10 flanked by SSR markers RM18193 - RM163 and RM171 - RM228, respectively.

CHROM	Monomorphic SSR (2021)	Polymorphic SSR (2020)	Polymorphic SSR (2021)	TOTAL Polymorphic SSR
1	4	9	2	11
2	6	5	0	5
3	5	3	1	4
4	6	2	0	2
5	18	4	4	8
6	4	3	1	4
7	4	2	1	3
8	4	3	2	5
9	0	4	0	4
10	0	2	0	2
11	0	5	0	5
12	32	0	5	5
Total	**83**	**42**	**16**	**58**

Table 2. List of SSR markers used in the verification of QTL associated to GLH resistance of PTRV Lead.

PROJECT 1

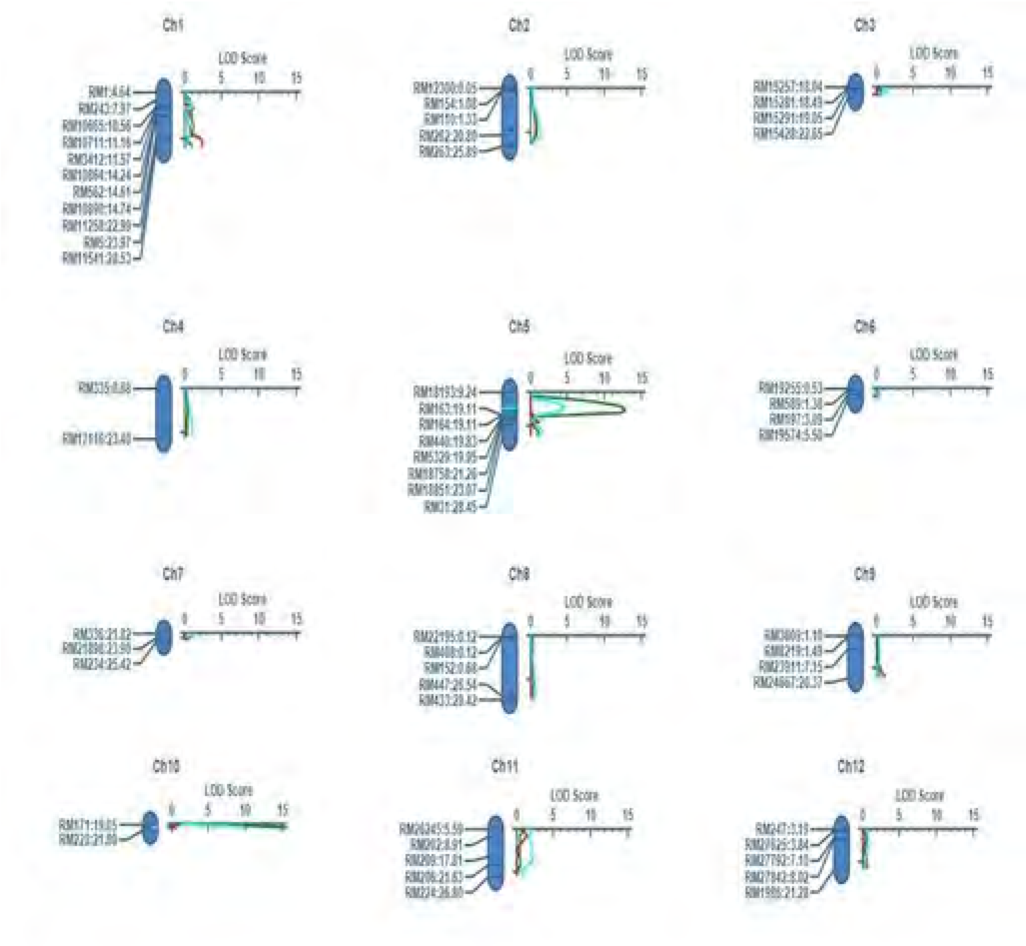

Figure 3. Location of the 58 polymorphic SSR markers used in the QTL analysis and the position of the identified QTL detected using the method Inclusive Composite Interval Mapping of QTL IciMapping software version 4.2.

Antibiosis	Chrom	Position	Left Marker	Right Marker	LOD	PVE (%)
Day_4	5	17.24	RM18193	RM163	12.8	3.6
Day_4	10	21.05	RM171	RM228	15.6	6.8
Day_5	5	16.24	RM18193	RM163	4.5	3.6
Day_5	10	21.05	RM171	RM228	9.9	6.6

Table 4. Updated QTL and linkage analysis showing consistent association of QTL regions at chromosomes 5 and 10 with the percent nymph mortality observed on the 4th and 5th days of the antibiosis test.

WHOLE GENOME SEQUENCING OF SELECTED RICE MUTANTS

Arlen A. dela Cruz, Teodora E. Mananghaya, Editha V. Evangelista, Mary Joyce G. Yapchiongco, Eleanor S. Avellanoza, Ma. Johna C. Duque, Christopher C. Cabusora, and Nenita V. Desamero

The whole genome DNA sequences of selected rice mutants and PTRV wild type, the high-quality DNA required to pass the quality assessment for whole-genome sequencing was achieved after several DNA extraction attempts. With the routine CTAB extraction method used in the laboratory, a high amount of RNA and debris contamination was observed. Despite the supplemental RNAse treatment, the high quality required for DNA samples for whole-genome sequencing was not achieved until the commercial Qiagen DNeasy® Plant Mini Kit was used. Table 4 shows the quality and quantity assessment report submitted by the DNA sequencing service provider. One of the four samples submitted, however, required further purification.

Table 5. Nanodrop and Agarose DNA quality assessment of DNA samples submitted for whole-genome sequencing.

Library Type	Sample Type	Amount (Qubit)	Volume	Concentration	Purity (NAnoDrop and Agarose Gel Assessment)
Plant Whole Genome Library (≤500bp)	Genomic DNA	≥200 ng	≥20 ul	≥10 ng/ul	OD 260/280 = 1.8-2.0; no degradation, no contamination
	Genomic DNA (PCR-free non-350bp	≥5 ng	≥20 ul	≥30 ng/ul	
	Genomic DNA (PCR-free 350bp	≥1.5 ng	≥20 ul	≥20 ng/ul	

FAS Apical conducted the whole-genome sequencing of rice while the 1st Base Nova LifeTech provided the basic bioinformatics analysis. Illumina raw reads were first removed from primer sequences using bbduk of the BBTools Packages. (https://jgi.doe.gov/data-and-tools/bbtools/). Next, QC-ed Illumina reads were error corrected using tadpole.sh. Finally, all reads were assembled de novo using Soapdenovo2 with the SOAPdenovo-63mer and scaffolding options. Table 5 shows the assembly statistics while Table 6 shows the annotation summary.

Table 5. Assembly statistics

# contigs (>= 0 bp)	81797	32277	29845	40373
# contigs (>= 1000 bp)	81797	32277	29845	40373
# contigs (>= 5000 bp)	13670	16016	15423	15806
# contigs (>= 10000 bp)	4161	9685	9706	8997
# contigs (>= 25000 bp)	419	2824	3126	2630
# contigs (>= 50000 bp)	54	469	637	468
Total length (>= 0 bp)	272237074	302496739	306529271	306699040
Total length (>= 1000 bp)	272237074	302496739	306529271	306699040
Total length (>= 5000 bp)	133733774	263497655	272250353	251325865
Total length (>= 10000 bp)	68148464	217959290	231245119	202932014
Total length (>= 25000 bp)	15108056	109701616	126533394	103603660
Total length (>= 50000 bp)	3245956	30230407	41908631	30386599
# contigs	81797	32277	29845	40373
Largest contig	108190	139352	164154	164243
Total length	272237074	302496739	306529271	306699040
GC (%)	42.43	42.47	42.56	42.55
N50	4873	18279	20549	16263
N75	2258	9002	10147	7044
L50	14154	4772	4306	5114
L75	35022	10626	9573	12224
# N's per 100 kbp	7788.93	4100.57	3907.74	4885.7

The gene prediction and annotation were conducted using the Funannotate pipeline (V1.8.9). Before gene prediction, repeats were soft masked using RepeatMasker 4.1.2-p1. Ab initio gene prediction was first made using GeneMark-ES v4.33 (Self training). Then, conserved orthologs were identified using BUSCO, and results from GeneMark-ES and BUSCO were used to train AUGUSTUS v3.3.3, snap, GlimmerHMM 3.0.4 for gene prediction. Protein evidence was obtained by aligning the predicted sequence to UniProtKB/Swiss-Prot/TrEMBLcurated database (v2021_04) using TBLASTN and Exonerate2.4.0. tRNAs were predicted with tRNAscanSE V2.0.9. Finally, videnceModeler was used to merge consensus gene models. The completeness of the assembled genome was evaluated using BUSCO v5 by comparing it with the poales_odb10 database with 4896 single-copy genes (https://busco.ezlab.org/list_of_lineages.html). Predicted coding sequences (CDS) were annotated through alignment to the UniProt/SwissProt protein database (v2021_04) using Diamond v2.0.13.151 and BlastP. Putative proteases and protease inhibitors were predicted using the MEROPS database. Protein families (PFAM version 35.0) and Gene Ontology (GO) terms were assigned with InterProScan v5.48-83.0 and aligned to the eggNOG 5 orthology database using emapper v2.1.3. SignalP v5.0 was used to annotate secretome; putative CAZymes were identified using HMMER v3.3 and family-specific hidden Markov model profiles of dbCAN2 meta server.

For variant calling, QC-ed unassembled reads were mapped to the reference genome (IR64_genome:https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7202035/) using Bowtie2 with local alignment type and medium sensitive settings. Then, variant calling was done using Geneious Prime using the standard translational code.

	Gal_ong	GXB	NSIC_Rc222	NSICRc222SM2
Assembly Size	272,237,074 bp	302,496,739 bp	306,529,271 bp	306,699,040 bp
Largest Scaffold	108,190 bp	139,352 bp	164,154 bp	164,243 bp
Average Scaffold	3,328 bp	9,372 bp	10,271 bp	7,597 bp
Num Scaffolds	81797	32277	29845	40373
Scaffold N50	4,873 bp	18,279 bp	20,549 bp	16,263 bp
Percent GC	39.13%	40.73%	40.90%	40.47%
Num Genes	36052	37562	38654	38056
Num Proteins	35955	37448	38548	37878
Num tRNA	97	114	106	178
Unique Proteins	11386	5796	5149	5317
Prots atleast 1 ortholog	23358	31056	32861	32042
Single-copy orthologs	7698	7698	7698	7698

Table 6. Annotation Summary

Two hundred seventy-five lines of NSIC Rc160 X Kutsiyam were screened for functional stay-green experiment in the screenhouse. These lines were clustered and divided into three groups based on their maturing characteristics; early maturing, mid-maturing, and late maturing. Each group was subjected to two treatments applied using the bucket method system to determine the stay-green capability in response to continuously waterlogged (CWL) and progressive drought (PDR). Water was maintained 2cm above the soil surface in CWL while in PDR while water was withheld at panicle initiation (PI) and allowed to dry up to 10% soil moisture content (SMC). SMC condition was maintained until harvest. SPAD value and Leaf Color Chart (LCC) were measured three times. The flag leaf was measured using a chlorophyll meter (SPAD-502, Minolta, Japan) from heading until the 14th day. Plant height and the number of tillers were measured and collected at 30DAT, 45DAT, 60DAT, and 80% maturity before harvesting. Shoots were gathered and harvested for the computation of biomass, yield, and other yield components (number and weight of filled and unfilled grains and 1,000 grain weight).

In the regions where the QTLs were detected, we selected lines with the Kutsiyam allele, the donor of the stay-green characteristic, and lines with a heterozygous allele (both parents› alleles). Compared with the check parent, NSIC Rc 160, we chose lines with higher grain weight and, simultaneously, a higher SPAD value (Figure 4).

Among 275 lines of NSIC Rc160 X Kutsiyam screened for the functional stay-green experiment, Line-50, Line-106, Line-42, Line-131, and Line-28 were identified as potential lines with functional stay-green traits (Figure 5). These lines had grain weight ranging from 9.52g to 16.65g, which was higher than the average of all lines (4.5g) and when compared with NSIC Rc 160, which only yielded 8.24g. On the other hand, the SPAD value of these lines ranged from 37.34 to 42.56, which is higher than the 37.08 value of the check parent and the 35.3 average value of all lines.

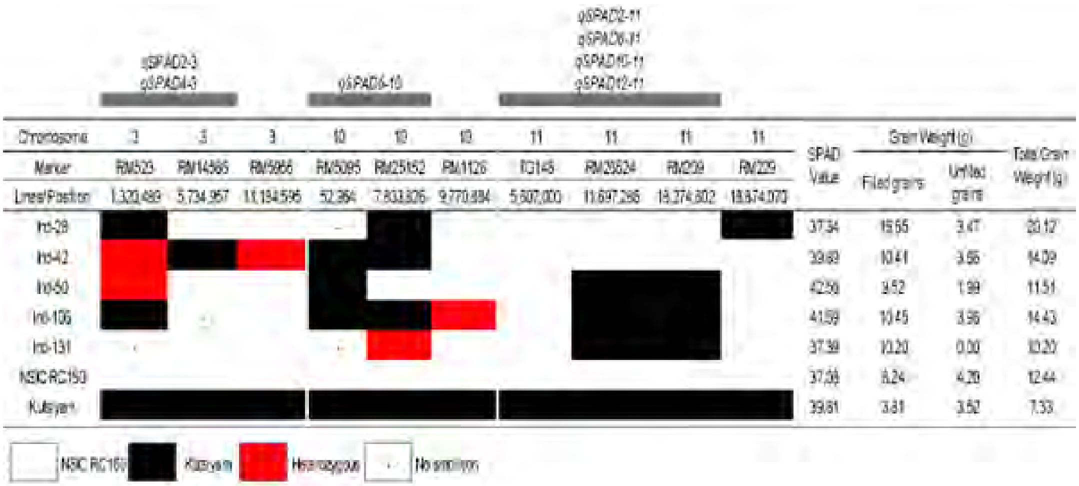

Figure 4. The QTLs identified in NSIC Rc160 X Kutsiyam that are associated with the functional stay-green trait and their corresponding effects contributed by either of the parents or both.

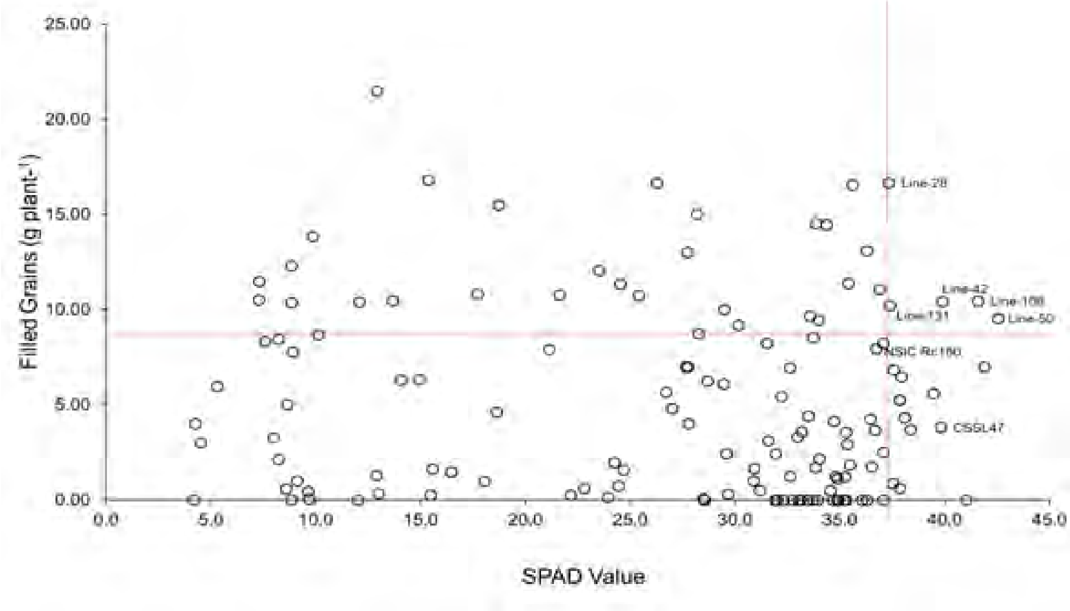

Figure 7. Relation of SPAD value and grain yield of the potential lines of NSIC Rc160 X Kutsiyam with functional stay-green characteristics under progressive drought, n=277.

PROJECT 2

Development of Genetically Engineered Microbes for Enhanced Rice Production and Industrial Uses

Reynante L. Ordonio

The development of genetically engineered microorganisms for enhanced rice production and industrial uses, specifically, biomass-degrading bacteria and bacteria expressing fusion protein for diagnostics, is one of PhilRice's innovations towards attaining a competitive rice economy through the use of biotechnology for food production, fuel, waste management, and value-added products. In addition, the project aimed to contribute to rice production waste management and support the breeding of tungro virus-resistant rice by developing technologies that efficiently convert waste to more valuable products and detect diseases to enhance yield and profitability.

- Three (3) guide RNA constructs targeting amylose biosynthesis, BLB, and Tungro resistance

- One (1) bacterium with potential Lignin-degrading ability

DEVELOPMENT OF A BACTERIUM WITH A LIGNIN-DEGRADING ACTIVITY

Reynante L. Ordonio, Jayvee A. Garcia, Mark Philip B. Castillo

We identified Lip1 gene coding for lignin-degrading protein lignin peroxidase isozyme. Information obtained include gene sequence (1,104 bp) and source microorganism (Trametes versicolor) (Figure 1).

GENE INFORMATION from the GenBank database
Gene symbol: LiP1
Gene description: lignin peroxidase isozyme
Locus tag: TRAVEDRAFT_43576
Gene type: protein coding
RNA name: lignin peroxidase isozyme
RefSeq status: PROVISIONAL
Organism: Trametes versicolor FP-101664 SS1 (strain: FP-101664 SS1) Lineage Eukaryota; Fungi; Dikarya; Basidiomycota; Agaricomycotina; Agaricomycetes; Polyporales; Trametes

PROJECT 2

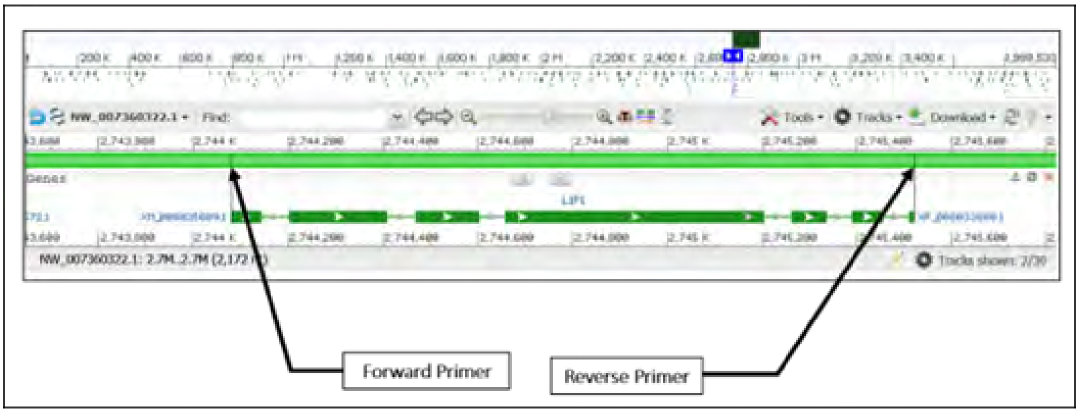

Figure 1. Site of LiP1 (lignin peroxidase isozyme) positioned at (bp 2,744,080-2,745,520) from the whole genome shotgun sequence (3,999,533 bp) of Trametes versicolor FP-101664 SS1 unplaced genomic scaffold TRAVEscaffold_2.
Source: *https://www.ncbi.nlm.nih.gov/nuccore/NW_007360322.1?report=graph*

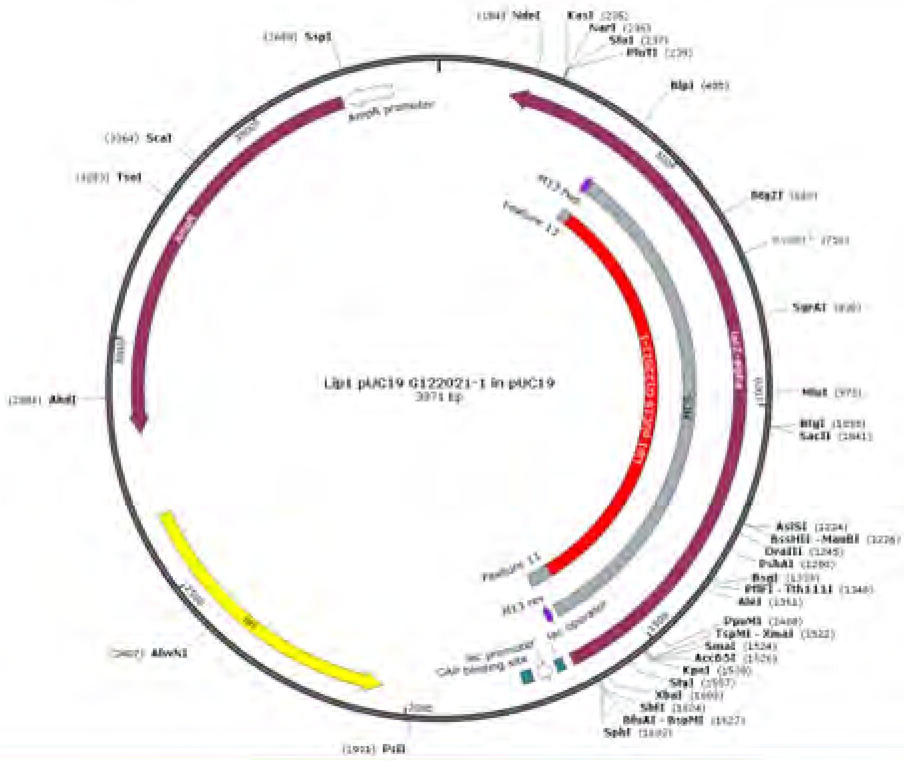

Figure 2. Map of the plasmid vector pUC19 inserted with the designed DNA sequence containing LiP1 gene coding sequence, linkers for digestion with restriction enzymes, additional cloning sites, and T7 terminator sequence.

BREEDING VIRUS-RESISTANT RICE BY EMPLOYING RECOMBINANT DNA TECHNOLOGY IN DEVELOPING DISEASE-DETECTION TECHNOLOGIES

Arlen A. dela Cruz, Ma. Johna C. Duque, Editha A. Evangelista, and Ladie Genesis B. Armecin

In 2021, apparently tungro-infected rice plants were collected from reported hotspot areas in Albay (Libon, Polangui, Oas, Tabaco), Camarines Sur (San Fernando), Nueva Ecija (Aliaga, General Natividad, Cabanatuan, and Munoz), Isabela (Burgos and Cubagan) and Laguna (Santa Maria, Calauan, Siniloan, Famy, Pagsanjan, and

	Tungro Hotspot Areas (2021,DS)		Tungro Hotspot Areas (2021, WS)
1	Zone 5, Libon Albay	1	San Miguel Burgos, Isabela
2	San Agustin, Libon Albay	2	Angan Casilla Cabagan, Isabela
3	West Carisac, Libon Albay	3	Cubag Cabagan, Isabela
4	Gabon, Polangui Albay	4	Roxas, Isabela
5	Kinali, Polangui Albay	5	Gabon Palangui, Albay
6	Balangibang, Polangui Albay	6	Zone 5, Libon, Albay
7	Apad, Polangui Albay	7	Centro Occidental Polangui, Albay
8	San Juan, OAS Albay	8	San Agustin, Libon, Albay
9	Busac Oas Albay	9	Kinali Polangui, Albay
10	Bongabong, Tabaco Albay	10	Busac, Oas, Albay
11	San Fernando Cam. Sur	11	Bongabon, Tabaco, Albay
12	Beberon, San Fernando, Cam. Sur	12	Santa Arcadia, Cabanatuan City, N.E
13	Betes Aliaga, NE	13	Balangkari Norte, Gen Natividad, NE
14	Betes Aliaga, NE	14	Talabutab, Gen Natividad NE
15	Balangkari Sur. Gen Natividad, NE	15	Talangka Sta Maria, Laguna
16	Balangkari Norte, Gen Natividad, NE	16	Dayap Calanan, Laguna
17	Mataas na Kahoy, Gen Natividad, NE	17	Acevida Siniloan, Laguna
18	PhilRice CES field	18	Batuhan Famy, Laguna
		19	Pinagsanjan Pagsanjan, Laguna
		20	Gatid Sta Cruz, Laguna
		21	San Fernando, Camsur

Table 2. Areas in Nueva Ecija, Camarines Sur, Nueva Ecija, Isabela, and Laguna verified with tungro cases in 2021.

Sta. Cruz (Table 2.)

The presence and quantity of tungro viruses in the plant samples were determined by ELISA, and the verified tungro-infected plants were used a virus transmission test with 40 tungro-susceptible TN1 seedlings. Figure 3 shows the percentage number of inoculated TN1 rice plants that were infected with RTBV-alone, RTSV-alone, RTBV+RTSV in the transmission test conducted during the DS. It should be noted that the tungro-infected plants, which were used as virus source plants in the transmission test, were collected at different growth stages (from maximum tillering to maturity) which might as well had affected the virus transmission efficiency. Based on the initial data, however, the tungro viruses from PhilRice CES were among the most virulent, together with those collected from Albay (Gabon, Polangui, Apad, Polagui, and San Juan, Oas). Most of the source plants were also infected with diseases other than tungro hence leaf samples used in the extraction of genetic materials for cloning and sequencing were taken from pooled TN1 plants used in

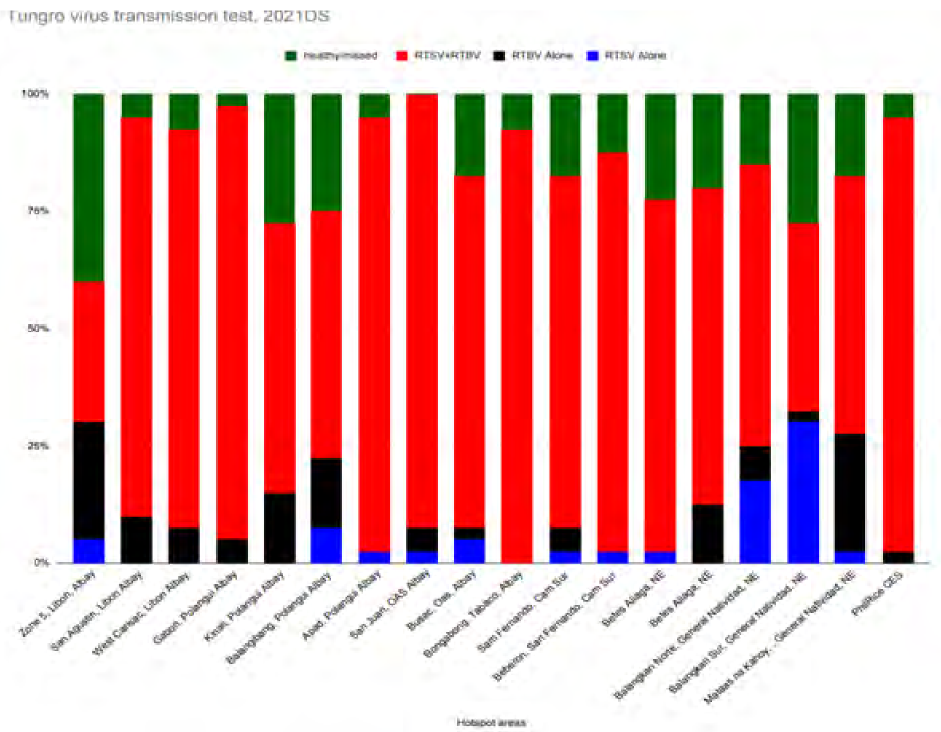

Figure 3. Percentage of TN1 plants with single or dual infection of RTBV and RTSV based on the virus transmission test with tungro-infected plants collected from reported tungro hotspot areas in Luzon in 2021 DS. n=40.

the transmission test.

The DNA and RNA samples (Figure 4) were extracted among inoculated TN1 plants in the transmission test. During the dry season (DS), genetic materials were extracted from pooled samples comprised of selected diseased plants showing obvious symptoms such as leaf discoloration and stunting. During the wet season (WS), however, sampling was done across all inoculated plants so as not to miss to represent the populations of less-virulent tungro virus strains,

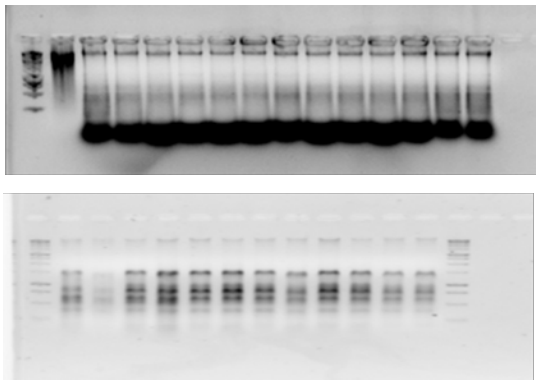

Figure 4. RNA (top) and DNA (bottom) samples isolated from tungro-infected rice plants and used as templates in the PCR-based amplification of the RTSV and RTBV coat protein genes (CP1, CP2, and CP3).

if there are.
The extracted DNA and cDNA synthesized were used to PCR-amplify the coat protein genes (CP1 CP2 and CP3) of RTBV and RTSV, respectively. The primers used are shown in Table 3. The amounts of MgCl2 (1mM, 1.5mM, and 2mM) and template (25-100ng), and PCR cycling annealing temperatures (50,55,60oC) were optimized for each primer. Figure 5 shows that in the amplification of the RTSV CP genes, 25ng cDNA only worked with CP3 primers CP3-1 and CP3-3. The RTSV CP1-1, CP1-2, and CP2-2 primers, on the other hand, showed amplification products with 75 ng cDNA template at 55 and 60oC annealing temperature (data not shown). Figure 6, on the other hand, shows the amplification of the RTBV CP1 gene with CP-n primer at 60oC annealing temperature. Two RTBV ORF primers were also included in the optimization but these are not intended for CP

Primers	Pair No.	Sequences (5' --> 3')
RTSV CP1-F	1	GCT GGT GAG ACA GTT ATT GT
RTSV CP1-R		CTG GCG AAT CGT ATG TTG
RTSV CP1-F	2	GGG GAT CCG CTG GCG AAA CGG TCA TTG
RTSV CP1-R		GGC TCG AGC TCA CTG GCC AAT CGT ATG
RTSV CP2-F	1	AGC GGG GGT TTC GAA GAG TCA
RTSV CP2-R		CTG TGC GGT CAG CAT CAC
RTSV CP2-F	2	GGG GAT CCA GCG GGG GTT TCG AAG AG
RTSV CP2-R		GGG CTC GAG CTC ACT GTG CGG TCA GCA CAT C
RTSV CP2-F	3	GGG CTC GAG CTC ACT GTG CGG TCA GCA CAT C
RTSV CP2-R		GGG AAT TCG ACT TTG GAA GAA GCC TAT C
RTSV CP3-F	1	GGT CTA GAC TCA TTG AGG ATC TGA CAC CGT AAA C
RTSV CP3-R		TTG GCA CCT CCG TCA CAC TCT TA
RTSV CP3-F	2	GAC TTT GGA AGA AGC CTA
RTSV CP3-R		TTG AGG ATC TGA CAC CGT
RTSV CP3-F	3	GGG GAT CCG ACT TTG GAA GAA GCC TA
RTSV CP3-R		GGC TCG AGC TCA TTG AGG ATC TGA CA

Table 3. Primers tested and optimized for PCR-amplification of RTBV and RTSV coat protein genes (CP1, CP2, and CP3)and CP3).

PROJECT 2

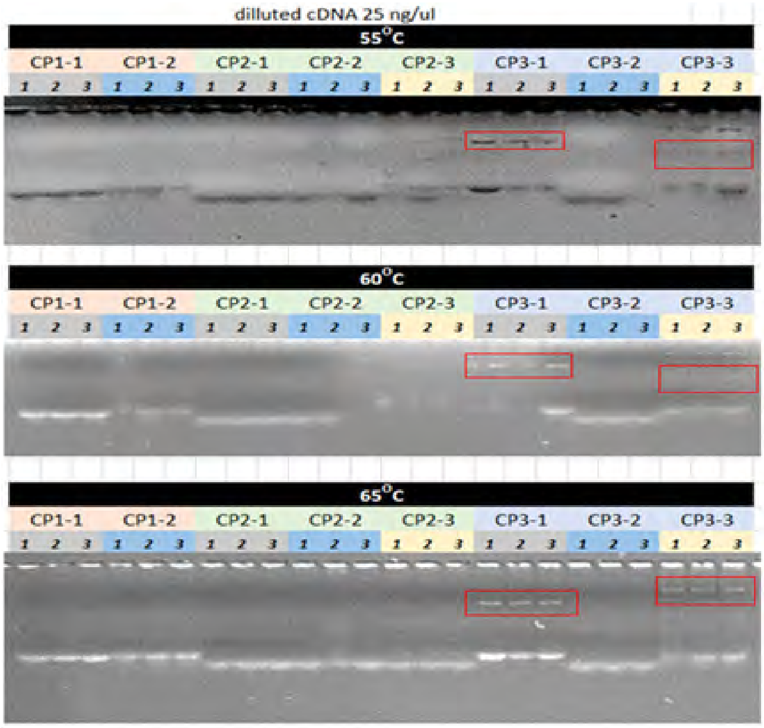

Figure 5. PCR-based amplification of RTSV CP genes at different annealing temperatures (55oC, 60oC, and 65oC) using 25ng cDNA template showing successful amplification of the CP3 gene with primers CP3-1 and CP3-3 at 65oC.

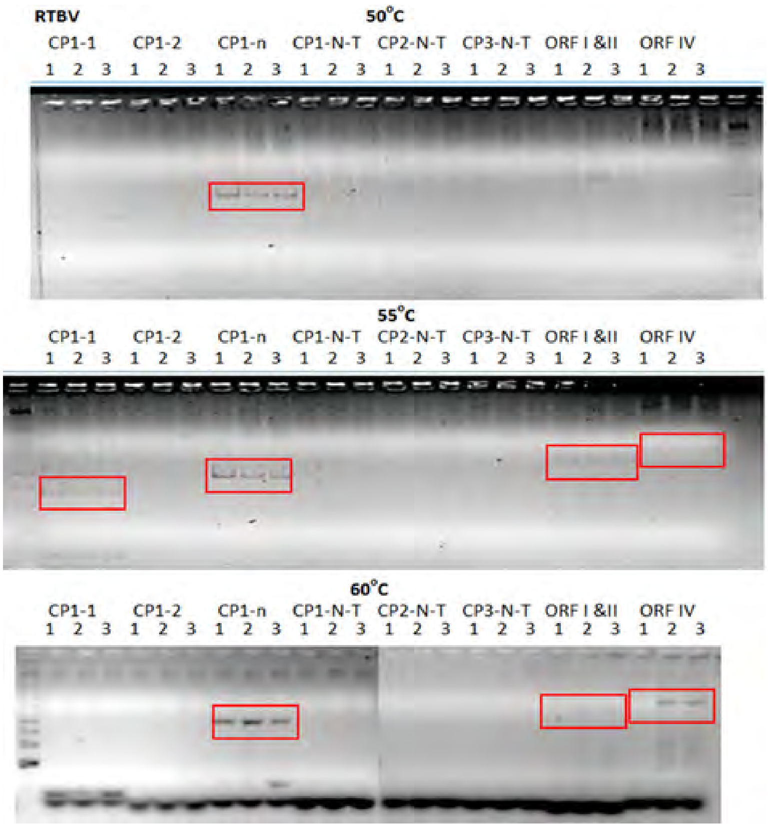

Figure 6. PCR-based amplification of RTBV CP genes at different annealing temperatures (50oC, 55oC and 60oC) showed successful amplification of the CP1 gene with primers CP1-n and CP1-1, including ORF I-II and ORF IV at 55oC and 60oC, respectively.

gene cloning but instead for the RTBV molecular diversity analysis. One primer per gene was selected and used in the PCR-based amplification of CP genes across RTSV cDNA or RTBV DNA samples representing the 2021 collection of tungro viruses from different hotspot areas with confirmed tungro cases (Figure 7). The amplified products were purified from agarose gels and cloned in pGEM T-easy vector. Recombinant bacteria carrying the different CP gene fragments were produced and these were sent for sequencing.

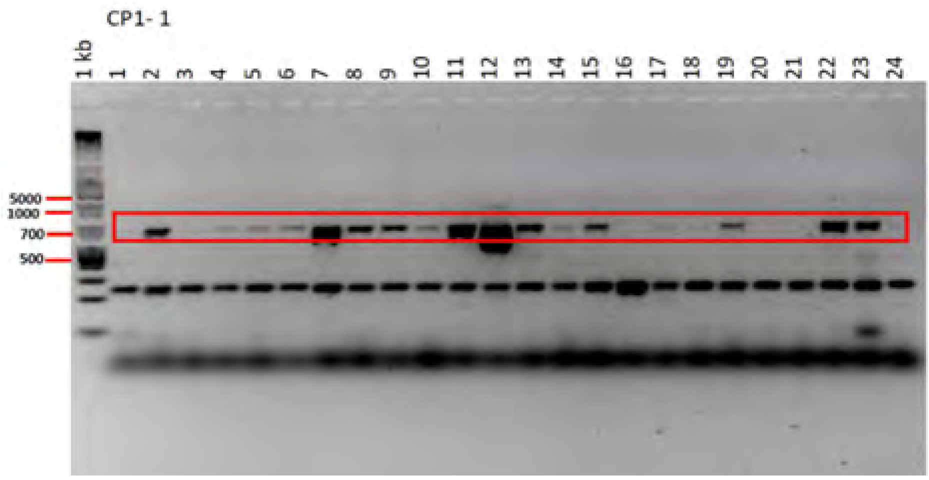

Figure7. PCR-based amplification with the selected CP1 primer (RTSV CP1-1) across RTSV cDNA samples representing the 2021 collection of tungro viruses from different hotspot areas with confirmed tungro cases.

Extracore Project 1

Development of provitamin A-enriched and disease-resistant rice lines through marker-assisted breeding

Reynante L. Ordonio

Golden Rice (GR) is a new type of rice that produces the pro-Vitamin A beta-carotene in the endosperm and has the potential to address the persistent and pervasive problem of Vitamin A deficiency in the country. Currently, as part of the continuous effort in the development of locally adapted GR, one of the six identified GR2 event "E" (GR2E) lines from the background of PSB Rc82 is used as the Golden Rice donor for the introgression of Golden Rice trait and Line 27 used as resistance donor for rice tungro virus (RTV) and bacterial leaf blight (BLB) into 6 high-yielding popular Philippine rice varieties through marker-assisted breeding approached.

- In 2021 dry season (21DS), we generated 55 entries composed of different advanced generations (45 BC4F1, 1 BC5F1, 9 BC3F1) with a total of 1627 individual plants (1334 BC4F1, 3 BC5F1, 290 BC3F1) derived from PSB Rc82-GR2E-38-4-27 as GR donor and Line 27 as disease resistance donor and NSIC Rc 222, NSIC Rc 238, NSIC Rc 300, NSIC Rc 216, and NSIC Rc 160 as recurrent parents, simultaneously, we included GR plant materials from IRRI composed of 195 plants from BC4F5 generations for disease evaluation of BLB and tungro. While, in 2021 wet season (21WS), we evaluated 72 entries (15 BC5F1, 35 BC4F2, 1 BC3F2,20 BC5F3, 4 BC3F1, 2 BC4F1,) with 313 individual plants (87 BC5F1, 167 BC4F2, 8 BC3F2, 49 BC5F3, 17 BC3F1,7 BC4F1,) derived from PSB Rc82-GR2E-384-27 and Line 27 as donor parent and NSIC Rc 238, NSIC Rc 300, NSIC Rc 160 as recurrent parents. All generated lines were phenotype for disease resistance (RTV and BLB) and genotyped using identified markers for GR, BLB and RTV for line selection. As of now, we have the most advanced lines from the background of NSIC Rc 300 at BC5F5 generation with 10 generated lines, 15 advanced lines (five each per background) for NSIC Rc 238, NSIC Rc 216 at BC4F4 and BC4F3 generation respectively and NSIC Rc 160 at BC4F3 generation.

Extracore Project 2

Improving Popular Released Rice Varieties Through Gene Editing

Reynante L. Ordonio, Trinidad C. Fernando, Ronalyn T. Miranda, Luilene A. Miranda, Mark Philip B. Castillo, Rhona Jane B. Fabros, and Marissa V. Romero

Rice breeding efforts in the Philippines are continuously being done to develop better quality and high yielding rice varieties. With the use of modern molecular biotechnology such as CRISPR/Cas9 or Cas12 systems, breeders now have a precise and faster alternative way of improving existing popular rice varieties. In this project, we use rice varieties that are being promoted through the Rice Competitiveness Enhancement Program (RCEP) of the Department of Agriculture.

- Initial steps were done to identify suitable recipient popular indica varieties that respond well to tissue culture to produce calli and floral dip to produce transformed seeds. Simultaneously, designing of appropriate constructs for gene editing by CRISPR-Cas technology was done, involving the following traits: rice tungro spherical virus (RTSV) resistance, bacterial leaf blight (BLB) resistance, improve aroma, and lower amylose content than the original. As a result, we identified 3 high yielding varieties (NSIC Rc 160, NSIC Rc 82 and NSIC Rc 512) out of 31 NSIC varieties tested, to be responsive to tissue culture (regeneration) and these were mass-produced for Agrobacterium inoculation. In addition, floral dip experiment using Agrobacterium was done, producing 1 putative transformant. Meanwhile, 15 primer/oligonucleotide pairs were designed for the guide RNA component of CRISPR-Cas9 vectors for gene editing: 5 for BLB resistance (OsSWEET 11, 13 and 14 genes); 3 for tungro virus resistance (tsv1 gene); and 7 for amylose content (Waxy gene). Vector construction for OsSWEET 14 gene is on-going.

Externally-funded Project 1:

Nutritionally Enhanced Rice: Finishing and Delivering Golden and High-iron & Zinc Rice Varieties

Reynante L. Ordonio, Ronalyn T. Miranda, Luilene A. Miranda, Mark Philip B. Castillo, Trinidad C. Fernando, and Rhoda Jane B. Fabros

The Healthier Rice project is composed of three teams, namely the Research-Breeding and Grain Quality and Nutrition team, Communication and Stakeholder Engagement team, and the Deployment team. Crop Biotech Center is involved in research breeding study and it has two main objectives: 1) to develop a pipeline GR2E varieties with different backgrounds (Golden Rice); and 2) Advance the product development and complete the biosafety regulatory dossier of high iron, high zinc rice (HIZR) in the Philippines. In 2021, 90 advanced GR2E introgression lines were generated consisting of 9 BC5F2, 13 BC5F1, 47 BC4F3, 12 BC4F2, 6 BC3F4, 1 BC3F3, and 2 BC2F2 in different backgrounds (derived from PSB Rc 82 GR2E:38-4-27 as donor parent and NSIC Rc 402, 358, 308, 300, and 298 as recurrent parent, all lines were screened using the GR markers BK29, BK30 and BK44). Also, a total of 39.28 kg of PSB Rc 82-GR2E and 36.11 kg of PSB Rc 82 were produced and transferred to the Grain Quality and Nutrition Team led by Dr. Marissa Romero. Recently the DA-BPI issued the biosafety permit for commercial propagation of Golden Rice "GR2E" on July 21, 2021. Upon the issuance of the permit the project produced a total of 130 kg of GR2E PSB Rc 82 (GR2E: 23-40-51) this will be used as additional seed source for seed production). Additionally, the project secured approval of NSIC Resolution No. 5 (A Resolution Adopting a Unified Policy by NSIC on Testing for Variety Registration of All Genetically Modified Crops on June 29, 2021, this exempts GR2E from NCT provided that the background used is an NSIC-approved variety. On the other hand, one season of HIZR CT were established and completed on October 5, 2021, and the proposal to DA-BPI for HIZR Field Trial was submitted on December 6, 2021.